

Module-3 (Assignment Day 5)

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.



```
C:\Users\dhairya mistry>ping 192.168.31.102

Pinging 192.168.31.102 with 32 bytes of data:
Reply from 192.168.31.102: bytes=32 time<1ms TTL=128
Reply from 192.168.31.102: bytes=32 time<1ms TTL=128
Reply from 192.168.31.102: bytes=32 time<1ms TTL=128
Reply from 192.168.31.102: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.31.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\dhairya mistry>|
```

```
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\dhairya mistry>ping 192.168.31.14

Pinging 192.168.31.14 with 32 bytes of data:
Reply from 192.168.31.14: bytes=32 time=339ms TTL=64
Reply from 192.168.31.14: bytes=32 time=36ms TTL=64
Reply from 192.168.31.14: bytes=32 time=38ms TTL=64
Reply from 192.168.31.14: bytes=32 time=155ms TTL=64

Ping statistics for 192.168.31.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 339ms, Average = 142ms

C:\Users\dhairya mistry>|
```

2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.



```
C:\Users\dhairya mistry>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4826:400::]
over a maximum of 30 hops:

  1    4 ms    3 ms    4 ms  jiofiber.local.html [2409:4090:1010:ee03:caa3:e8ff:fede:6df8]
  2    *      *      *      Request timed out.
  3   31 ms   18 ms   12 ms  2405:203:400:100:172:31:3:164
  4   17 ms   16 ms   65 ms  2405:200:801:b00::f3a
  5  1300 ms  147 ms   28 ms  2405:200:809:1510:61::5
  6    *      *      *      Request timed out.
  7   27 ms   42 ms   71 ms  2405:200:801:b00::dfa
  8   117 ms  231 ms   46 ms  2405:200:801:b00::df9
  9    *      231 ms  317 ms  2001:4860:4826:400::

Trace complete.
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.



```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhairya mistry>ping 192.168.31.102

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```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.



```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\dhairya mistry>arp -a

Interface: 192.168.31.102 --- 0x2
Internet Address      Physical Address      Type
192.168.31.1          c8-a3-e8-de-6d-f8    dynamic
192.168.31.65         20-51-f5-92-ab-4c    dynamic
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

ARP Table

Internet Address (IP)	Physical Address (MAC)	Type
192.168.31.1	C8-A3-E8-DE-6D-F8	Dynamic
192.168.31.65	20-51-F5-92-AB-4C	Dynamic
224.0.0.22	01-00-5E-00-00-16	Static
224.0.0.251	01-00-5E-00-00-FB	Static
224.0.0.252	01-00-5E-00-00-FC	Static
255.255.255.255	FF-FF-FF-FF-FF-FF	Static

Interface IP: 192.168.31.102

Router/Gateway: 192.168.31.1

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.



Short Network Testing Report

Objective: To set up a network and check whether devices could communicate properly.

Steps:

1. Connected the computer to the same Wi-Fi network as the other device.
2. Checked the IP address using ipconfig.
3. Used arp -a to view connected devices and their MAC addresses.
4. Tested connectivity using the ping command, for example: ping 192.168.31.1.
5. Used tracert to check the path taken by packets: tracert 192.168.31.1.
6. Checked the results and confirmed whether packets were successfully transmitted.

Results

The computer had IP address 192.168.31.102, and the router was 192.168.31.1. The ARP table showed the router and another device (192.168.31.65). Ping was used to test connectivity, while tracert showed the network path. No major network issue was observed.

```
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